

* Rank 0 tensor.

$$[Q]$$

$$Q = Q_s^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} 0 & Q_y & 0 \end{bmatrix}$$

$$Q_y = Q_{py}^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & Q_{xz} \\ 0 & Q_{yy} & 0 \\ Q_{xz} & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = -Q_{du}^{(1)} + Q_{dv}^{(1)} + Q_s^{(1)}$$

$$Q_{xz} = Q_{dxz}^{(1)}$$

$$Q_{yy} = -Q_{du}^{(1)} - Q_{dv}^{(1)} + Q_s^{(1)}$$

$$Q_{zz} = 2Q_{du}^{(1)} + Q_s^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & Q_{xz} \\ 0 & 0 & 0 \\ -Q_{xz} & 0 & 0 \end{bmatrix}$$

$$Q_{xz} = -G_{py}^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & Q_{xxy} & 0 \\ 0 & Q_{yyy} & 0 \\ 0 & Q_{zzy} & 0 \\ Q_{yzx} & 0 & Q_{zzy} \\ 0 & Q_{yzx} & 0 \\ Q_{xxy} & 0 & Q_{yzx} \end{bmatrix}$$

$$Q_{xxy} = -Q_{fay}^{(1)} - Q_{fby}^{(1)} + Q_{py}^{(1)}$$

$$Q_{yyy} = 2Q_{fay}^{(1)} + 3Q_{py}^{(1)}$$

$$Q_{zzy} = -Q_{fay}^{(1)} + Q_{fby}^{(1)} + Q_{py}^{(1)}$$

$$Q_{yzx} = Q_{f3}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & Q_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & Q_{zzy} & 0 \\ Q_{yzx} & 0 & -\frac{Q_{zzy}}{2} \\ 0 & Q_{zxy} & 0 \\ -\frac{Q_{xxy}}{2} & 0 & -Q_{yzx} - Q_{zxy} \end{bmatrix}$$

$$Q_{xxy} = -2G_{dxz}^{(1)} + 2Q_{py}^{(2)}$$

$$Q_{zzy} = 2G_{dxz}^{(1)} + 2Q_{py}^{(2)}$$

$$Q_{yzx} = 3G_{du}^{(1)} + G_{dv}^{(1)}$$

$$\frac{Q_{zzy}}{2} = G_{dxz}^{(1)} + Q_{py}^{(2)}$$

$$Q_{zxy} = -3G_{du}^{(1)} + G_{dv}^{(1)}$$

$$\frac{Q_{xxy}}{2} = -G_{dxz}^{(1)} + Q_{py}^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} Q_{yzx} & 0 & Q_{yzz} \\ 0 & Q_{zxy} & 0 \\ Q_{xyx} & 0 & Q_{xyz} \end{bmatrix}$$

$$Q_{yzx} = -G_{du}^{(2)} + G_{dv}^{(2)} + G_s^{(1)}$$

$$Q_{yzz} = G_{dxz}^{(2)} - Q_{py}^{(3)}$$

$$Q_{zxy} = -G_{du}^{(2)} - G_{dv}^{(2)} + G_s^{(1)}$$

$$Q_{xyx} = G_{dxz}^{(2)} + Q_{py}^{(3)}$$

$$Q_{xyz} = 2G_{du}^{(2)} + G_s^{(1)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 \\ Q_{xxyy} & Q_{yyyy} & Q_{yyzz} & 0 & Q_{yyzx} & 0 \\ Q_{xxzz} & Q_{yyzz} & Q_{zzzz} & 0 & Q_{zzzx} & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} \\ Q_{xxzx} & Q_{yyzx} & Q_{zzzx} & 0 & Q_{zxzx} & 0 \\ 0 & 0 & 0 & Q_{yzxy} & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{xxxx} = -2Q_{du}^{(1)} - 4Q_{du}^{(2)} + 2Q_{dv}^{(1)} + 4Q_{dv}^{(2)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} + 2Q_g^{(1)} + Q_s^{(1)} + 2Q_s^{(2)}$$

$$Q_{xxyy} = -2Q_{du}^{(1)} + 2Q_{gu}^{(1)} - Q_g^{(1)} + Q_s^{(1)}$$

$$Q_{xxzz} = Q_{du}^{(1)} + Q_{dv}^{(1)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(1)}$$

$$Q_{xxzx} = Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} - Q_{gay}^{(1)} - Q_{gby}^{(1)}$$

$$Q_{yyyy} = -2Q_{du}^{(1)} - 4Q_{du}^{(2)} - 2Q_{dv}^{(1)} - 4Q_{dv}^{(2)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} + 2Q_g^{(1)} + Q_s^{(1)} + 2Q_s^{(2)}$$

$$Q_{yyzz} = Q_{du}^{(1)} - Q_{dv}^{(1)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(1)}$$

$$Q_{yyzx} = Q_{dxz}^{(1)} + 2Q_{gby}^{(1)}$$

$$Q_{zzzz} = 4Q_{du}^{(1)} + 8Q_{du}^{(2)} + 2Q_{gu}^{(1)} + 2Q_g^{(1)} + Q_s^{(1)} + 2Q_s^{(2)}$$

$$Q_{zzzx} = Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} + Q_{gay}^{(1)} - Q_{gby}^{(1)}$$

$$Q_{yzyz} = Q_{du}^{(2)} - Q_{dv}^{(2)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(2)}$$

$$Q_{yzxy} = Q_{dxz}^{(2)} + 2Q_{gby}^{(1)}$$

$$Q_{zxzx} = Q_{du}^{(2)} + Q_{dv}^{(2)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(2)}$$

$$Q_{xyxy} = -2Q_{du}^{(2)} + 2Q_{gu}^{(1)} - Q_g^{(1)} + Q_s^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 \\ -Q_{xxyy} & 0 & Q_{yyzz} & 0 & Q_{yyzx} & 0 \\ -Q_{xxzz} & -Q_{yyzz} & 0 & 0 & Q_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{yzxy} \\ -Q_{xxzx} & -Q_{yyzx} & -Q_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -Q_{yzxy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
Q_{xxyy} &= 4G_{f3}^{(1)} - 2Q_{dv}^{(3)} \\
Q_{xxzz} &= -4G_{f3}^{(1)} + 3Q_{du}^{(3)} - Q_{dv}^{(3)} \\
Q_{xxxz} &= 2G_{fay}^{(1)} + 2G_{fby}^{(1)} - 2G_{py}^{(1)} + Q_{dxz}^{(3)} \\
Q_{yyzz} &= 4G_{f3}^{(1)} + 3Q_{du}^{(3)} + Q_{dv}^{(3)} \\
Q_{yyyz} &= -4G_{fby}^{(1)} + Q_{dxz}^{(3)} \\
Q_{zzzz} &= -2G_{fay}^{(1)} + 2G_{fby}^{(1)} + 2G_{py}^{(1)} + Q_{dxz}^{(3)} \\
Q_{yzxy} &= 4G_{fay}^{(1)} + G_{py}^{(1)}
\end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & Q_{yzxy} \\ 0 & Q_{zzzx} & 0 \\ Q_{yzxy} & 0 & Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
Q_{yzyz} &= -2Q_{du}^{(4)} + 2Q_{dv}^{(4)} + Q_s^{(3)} \\
Q_{yzxy} &= 2Q_{dxz}^{(4)} \\
Q_{zzzx} &= -2Q_{du}^{(4)} - 2Q_{dv}^{(4)} + Q_s^{(3)} \\
Q_{xyxy} &= 4Q_{du}^{(4)} + Q_s^{(3)}
\end{aligned}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & Q_{yzxy} \\ 0 & 0 & 0 \\ -Q_{yzxy} & 0 & 0 \end{bmatrix}$$

$$Q_{yzxy} = G_{py}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & Q_{xxzx} & 0 \\ 0 & Q_{yyzx} & 0 \\ 0 & Q_{zzzx} & 0 \\ Q_{yzyz} & 0 & Q_{yzxy} \\ 0 & Q_{zzzx} & 0 \\ Q_{xyyz} & 0 & Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
Q_{xxzx} &= -G_{fay}^{(2)} - G_{fby}^{(2)} + G_{py}^{(3)} + 2Q_{dxz}^{(5)} \\
Q_{yyzx} &= 2G_{fay}^{(2)} + G_{py}^{(3)} + 2G_{py}^{(4)} \\
Q_{zzzx} &= -G_{fay}^{(2)} + G_{fby}^{(2)} + G_{py}^{(3)} - 2Q_{dxz}^{(5)} \\
Q_{yzyz} &= G_{f3}^{(2)} - 3Q_{du}^{(5)} - Q_{dv}^{(5)} \\
Q_{yzxy} &= -G_{fay}^{(2)} + G_{fby}^{(2)} + G_{py}^{(4)} + Q_{dxz}^{(5)} \\
Q_{xxzx} &= G_{f3}^{(2)} + 3Q_{du}^{(5)} - Q_{dv}^{(5)} \\
Q_{xyyz} &= -G_{fay}^{(2)} - G_{fby}^{(2)} + G_{py}^{(4)} - Q_{dxz}^{(5)} \\
Q_{xyxy} &= G_{f3}^{(2)} + 2Q_{dv}^{(5)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} \\ Q_{zzxx} & Q_{zxyy} & Q_{zzzz} & 0 & Q_{zzzx} & 0 \\ 0 & 0 & 0 & Q_{xyyz} & 0 & Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
Q_{yzyz} &= G_{f3}^{(3)} - 3Q_{du}^{(6)} - Q_{dv}^{(6)} \\
Q_{yzxy} &= -G_{fay}^{(3)} - G_{fby}^{(3)} + G_{py}^{(6)} - Q_{dxz}^{(6)} \\
Q_{zxxx} &= -G_{fay}^{(3)} - G_{fby}^{(3)} + G_{py}^{(5)} + 2Q_{dxz}^{(6)} \\
Q_{zxyy} &= 2G_{fay}^{(3)} + G_{py}^{(5)} + 2G_{py}^{(6)} \\
Q_{zxzz} &= -G_{fay}^{(3)} + G_{fby}^{(3)} + G_{py}^{(5)} - 2Q_{dxz}^{(6)} \\
Q_{zzxx} &= G_{f3}^{(3)} + 3Q_{du}^{(6)} - Q_{dv}^{(6)} \\
Q_{xyyz} &= -G_{fay}^{(3)} + G_{fby}^{(3)} + G_{py}^{(6)} + Q_{dxz}^{(6)} \\
Q_{xyxy} &= G_{f3}^{(3)} + 2Q_{dv}^{(6)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 & 0 & Q_{xxxz} & 0 \\
Q_{yyxx} & Q_{yyyy} & Q_{yyzz} & 0 & Q_{yyzx} & 0 & 0 & Q_{yyxz} & 0 \\
Q_{zzxx} & Q_{zzyy} & Q_{zzzz} & 0 & Q_{zzzx} & 0 & 0 & Q_{zzxz} & 0 \\
0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} & Q_{yzzy} & 0 & Q_{yzyx} \\
Q_{zxxx} & Q_{zxxy} & Q_{zxzz} & 0 & Q_{zxzx} & 0 & 0 & Q_{zxxz} & 0 \\
0 & 0 & 0 & Q_{xyyz} & 0 & Q_{xyxy} & Q_{xyzy} & 0 & Q_{xyyx}
\end{bmatrix}$$

$$\begin{aligned}
Q_{xxxx} &= -2Q_{du}^{(1)} - 4Q_{du}^{(2)} + 2Q_{dv}^{(1)} + 4Q_{dv}^{(2)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} + 2Q_g^{(1)} + Q_s^{(1)} + 2Q_s^{(2)} \\
Q_{xxyy} &= 4G_{f3}^{(1)} - 2Q_{du}^{(1)} - 2Q_{dv}^{(3)} + 2Q_{gu}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{xxzz} &= -4G_{f3}^{(1)} + Q_{du}^{(1)} + 3Q_{du}^{(3)} + Q_{dv}^{(1)} - Q_{dv}^{(3)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{xxzx} &= 2G_{fay}^{(1)} - G_{fay}^{(2)} + 2G_{fby}^{(1)} - G_{fby}^{(2)} - 2G_{py}^{(1)} + G_{py}^{(3)} + Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} + Q_{dxz}^{(3)} + 2Q_{dxz}^{(5)} - Q_{gay}^{(1)} - Q_{gby}^{(1)} \\
Q_{xxzx} &= 2G_{fay}^{(1)} + G_{fay}^{(2)} + 2G_{fby}^{(1)} + G_{fby}^{(2)} - 2G_{py}^{(1)} - G_{py}^{(3)} + Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} + Q_{dxz}^{(3)} - 2Q_{dxz}^{(5)} - Q_{gay}^{(1)} - Q_{gby}^{(1)} \\
Q_{yyxx} &= -4G_{f3}^{(1)} - 2Q_{du}^{(1)} + 2Q_{dv}^{(3)} + 2Q_{gu}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{yyyy} &= -2Q_{du}^{(1)} - 4Q_{du}^{(2)} - 2Q_{dv}^{(1)} - 4Q_{dv}^{(2)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} + 2Q_g^{(1)} + Q_s^{(1)} + 2Q_s^{(2)} \\
Q_{yyzz} &= 4G_{f3}^{(1)} + Q_{du}^{(1)} + 3Q_{du}^{(3)} - Q_{dv}^{(1)} + Q_{dv}^{(3)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{yyzx} &= 2G_{fay}^{(2)} - 4G_{fby}^{(1)} + G_{py}^{(3)} + 2G_{py}^{(4)} + Q_{dxz}^{(1)} + Q_{dxz}^{(3)} + 2Q_{gby}^{(1)} \\
Q_{yyxz} &= -2G_{fay}^{(2)} - 4G_{fby}^{(1)} - G_{py}^{(3)} - 2G_{py}^{(4)} + Q_{dxz}^{(1)} + Q_{dxz}^{(3)} + 2Q_{gby}^{(1)} \\
Q_{zzxx} &= 4G_{f3}^{(1)} + Q_{du}^{(1)} - 3Q_{du}^{(3)} + Q_{dv}^{(1)} + Q_{dv}^{(3)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{zzyy} &= -4G_{f3}^{(1)} + Q_{du}^{(1)} - 3Q_{du}^{(3)} - Q_{dv}^{(1)} - Q_{dv}^{(3)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{zzzz} &= 4Q_{du}^{(1)} + 8Q_{du}^{(2)} + 2Q_{gu}^{(1)} + 2Q_g^{(1)} + Q_s^{(1)} + 2Q_s^{(2)} \\
Q_{zzzx} &= -2G_{fay}^{(1)} - G_{fay}^{(2)} + 2G_{fby}^{(1)} + G_{fby}^{(2)} + 2G_{py}^{(1)} + G_{py}^{(3)} + Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} + Q_{dxz}^{(3)} - 2Q_{dxz}^{(5)} + Q_{gay}^{(1)} - Q_{gby}^{(1)} \\
Q_{zzzx} &= -2G_{fay}^{(1)} + G_{fay}^{(2)} + 2G_{fby}^{(1)} - G_{fby}^{(2)} + 2G_{py}^{(1)} - G_{py}^{(3)} + Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} + Q_{dxz}^{(3)} + 2Q_{dxz}^{(5)} + Q_{gay}^{(1)} - Q_{gby}^{(1)} \\
Q_{yzyz} &= G_{f3}^{(2)} + Q_{du}^{(2)} - 3Q_{du}^{(5)} - Q_{dv}^{(2)} - Q_{dv}^{(5)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(2)} \\
Q_{yzxy} &= 4G_{fay}^{(1)} - G_{fay}^{(2)} + G_{fby}^{(2)} + G_{py}^{(1)} + G_{py}^{(4)} + Q_{dxz}^{(2)} + Q_{dxz}^{(5)} + 2Q_{gby}^{(1)} \\
Q_{yzzy} &= -G_{f3}^{(2)} + Q_{du}^{(2)} + 3Q_{du}^{(5)} - Q_{dv}^{(2)} + Q_{dv}^{(5)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(2)} \\
Q_{zyyx} &= 4G_{fay}^{(1)} + G_{fay}^{(2)} - G_{fby}^{(2)} + G_{py}^{(1)} - G_{py}^{(4)} + Q_{dxz}^{(2)} - Q_{dxz}^{(5)} + 2Q_{gby}^{(1)} \\
Q_{zxxx} &= -2G_{fay}^{(1)} - 2G_{fby}^{(1)} + 2G_{py}^{(1)} + Q_{dxz}^{(2)} + 2Q_{dxz}^{(2)} - Q_{dxz}^{(3)} - Q_{gay}^{(1)} - Q_{gby}^{(1)} \\
Q_{zxyy} &= 4G_{fby}^{(1)} + Q_{dxz}^{(1)} - Q_{dxz}^{(3)} + 2Q_{gby}^{(1)} \\
Q_{zxxx} &= 2G_{fay}^{(1)} - 2G_{fby}^{(1)} - 2G_{py}^{(1)} + Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} - Q_{dxz}^{(3)} + Q_{gay}^{(1)} - Q_{gby}^{(1)} \\
Q_{zxxx} &= G_{f3}^{(2)} + Q_{du}^{(2)} + 3Q_{du}^{(5)} + Q_{dv}^{(2)} - Q_{dv}^{(5)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(2)} \\
Q_{zxxx} &= -G_{f3}^{(2)} + Q_{du}^{(2)} - 3Q_{du}^{(5)} + Q_{dv}^{(2)} + Q_{dv}^{(5)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(2)} \\
Q_{xyyz} &= -4G_{fay}^{(1)} - G_{fay}^{(2)} - G_{fby}^{(2)} - G_{py}^{(1)} + G_{py}^{(4)} + Q_{dxz}^{(2)} - Q_{dxz}^{(5)} + 2Q_{gby}^{(1)} \\
Q_{xyxy} &= G_{f3}^{(2)} - 2Q_{du}^{(2)} + 2Q_{dv}^{(5)} + 2Q_{gu}^{(1)} - Q_g^{(1)} + Q_s^{(2)} \\
Q_{xyzy} &= -4G_{fay}^{(1)} + G_{fay}^{(2)} + G_{fby}^{(2)} - G_{py}^{(1)} - G_{py}^{(4)} + Q_{dxz}^{(2)} + Q_{dxz}^{(5)} + 2Q_{gby}^{(1)} \\
Q_{xyyx} &= -G_{f3}^{(2)} - 2Q_{du}^{(2)} - 2Q_{dv}^{(5)} + 2Q_{gu}^{(1)} - Q_g^{(1)} + Q_s^{(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix}
0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} & Q_{yzzy} & 0 & Q_{yzyx} \\
Q_{zxxx} & Q_{zxyy} & Q_{zxxx} & 0 & Q_{zxxx} & 0 & 0 & Q_{zxxx} & 0 \\
0 & 0 & 0 & Q_{xyyz} & 0 & Q_{xyxy} & Q_{xyzy} & 0 & Q_{xyyx}
\end{bmatrix}$$

$$\begin{aligned}
Q_{yzyz} &= G_{f3}^{(3)} - 2Q_{du}^{(4)} - 3Q_{du}^{(6)} + 2Q_{dv}^{(4)} - Q_{dv}^{(6)} + Q_s^{(3)} \\
Q_{yzxy} &= -G_{fay}^{(3)} - G_{fby}^{(3)} + G_{py}^{(2)} + G_{py}^{(6)} + 2Q_{dxz}^{(4)} - Q_{dxz}^{(6)} \\
Q_{yzzz} &= G_{f3}^{(3)} + 2Q_{du}^{(4)} - 3Q_{du}^{(6)} - 2Q_{dv}^{(4)} - Q_{dv}^{(6)} - Q_s^{(3)} \\
Q_{yzyx} &= -G_{fay}^{(3)} - G_{fby}^{(3)} - G_{py}^{(2)} + G_{py}^{(6)} - 2Q_{dxz}^{(4)} - Q_{dxz}^{(6)} \\
Q_{zxxx} &= -G_{fay}^{(3)} - G_{fby}^{(3)} + G_{py}^{(5)} + 2Q_{dxz}^{(6)} \\
Q_{zxyy} &= 2G_{fay}^{(3)} + G_{py}^{(5)} + 2G_{py}^{(6)} \\
Q_{zxxz} &= -G_{fay}^{(3)} + G_{fby}^{(3)} + G_{py}^{(5)} - 2Q_{dxz}^{(6)} \\
Q_{zxxz} &= G_{f3}^{(3)} - 2Q_{du}^{(4)} + 3Q_{du}^{(6)} - 2Q_{dv}^{(4)} - Q_{dv}^{(6)} + Q_s^{(3)} \\
Q_{zxxz} &= G_{f3}^{(3)} + 2Q_{du}^{(4)} + 3Q_{du}^{(6)} + 2Q_{dv}^{(4)} - Q_{dv}^{(6)} - Q_s^{(3)} \\
Q_{xyyz} &= -G_{fay}^{(3)} + G_{fby}^{(3)} - G_{py}^{(2)} + G_{py}^{(6)} + 2Q_{dxz}^{(4)} + Q_{dxz}^{(6)} \\
Q_{xyxy} &= G_{f3}^{(3)} + 4Q_{du}^{(4)} + 2Q_{dv}^{(6)} + Q_s^{(3)} \\
Q_{xyzy} &= -G_{fay}^{(3)} + G_{fby}^{(3)} + G_{py}^{(2)} + G_{py}^{(6)} - 2Q_{dxz}^{(4)} + Q_{dxz}^{(6)} \\
Q_{xyyx} &= G_{f3}^{(3)} - 4Q_{du}^{(4)} + 2Q_{dv}^{(6)} - Q_s^{(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
Q_{xxxx} & 0 & Q_{xxxx} \\
0 & Q_{yyyy} & 0 \\
Q_{zzzx} & 0 & Q_{zzzx} \\
Q_{yyzx} & 0 & Q_{yyzx} \\
Q_{zzxx} & 0 & Q_{zzxx} \\
0 & Q_{xxyy} & 0 \\
0 & Q_{yzyy} & 0 \\
Q_{zxzx} & 0 & Q_{zxzx} \\
Q_{xxyy} + Q_{yyzz} - Q_{yzyy} - Q_{zxxz} + Q_{zzxx} & 0 & Q_{xxyy} \\
0 & Q_{xyyz} + \frac{Q_{zzzz}}{2} - \frac{Q_{zzzx}}{2} & 0
\end{bmatrix}$$

$$\begin{aligned}
Q_{xxxx} &= -2Q_{du}^{(1)} - 4Q_{du}^{(2)} + 2Q_{dv}^{(1)} + 4Q_{dv}^{(2)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} + 2Q_g^{(1)} + 3Q_s^{(1)} \\
Q_{xxzx} &= Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} + Q_{dxz}^{(3)} - 2Q_{dxz}^{(5)} - Q_{gay}^{(1)} - Q_{gby}^{(1)} \\
Q_{yyyy} &= -2Q_{du}^{(1)} - 4Q_{du}^{(2)} - 2Q_{dv}^{(1)} - 4Q_{dv}^{(2)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} + 2Q_g^{(1)} + 3Q_s^{(1)} \\
Q_{zzzx} &= Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} + Q_{dxz}^{(3)} - 2Q_{dxz}^{(5)} + Q_{gay}^{(1)} - Q_{gby}^{(1)} \\
Q_{zzzz} &= 4Q_{du}^{(1)} + 8Q_{du}^{(2)} + 2Q_{gu}^{(1)} + 2Q_g^{(1)} + 3Q_s^{(1)} \\
Q_{yyzx} &= Q_{dxz}^{(1)} + Q_{dxz}^{(3)} + 2Q_{gby}^{(1)} \\
Q_{yzyz} &= Q_{du}^{(1)} + 3Q_{du}^{(3)} - Q_{dv}^{(1)} + Q_{dv}^{(3)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{zzxx} &= Q_{du}^{(1)} - 3Q_{du}^{(3)} + Q_{dv}^{(1)} + Q_{dv}^{(3)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{zzxz} &= Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} + Q_{dxz}^{(3)} + 2Q_{dxz}^{(5)} + Q_{gay}^{(1)} - Q_{gby}^{(1)} \\
Q_{xxyy} &= -2Q_{du}^{(1)} - 2Q_{dv}^{(3)} + 2Q_{gu}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{yzyy} &= Q_{du}^{(2)} + 3Q_{du}^{(5)} - Q_{dv}^{(2)} + Q_{dv}^{(5)} - Q_{gu}^{(1)} + Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{zxxx} &= Q_{dxz}^{(1)} + 2Q_{dxz}^{(2)} - Q_{dxz}^{(3)} - Q_{gay}^{(1)} - Q_{gby}^{(1)} \\
Q_{zxxz} &= Q_{du}^{(2)} - 3Q_{du}^{(5)} + Q_{dv}^{(2)} + Q_{dv}^{(5)} - Q_{gu}^{(1)} - Q_{gv}^{(1)} - Q_g^{(1)} + Q_s^{(1)} \\
Q_{xyyz} &= Q_{dxz}^{(2)} - Q_{dxz}^{(5)} + 2Q_{gby}^{(1)}
\end{aligned}$$